

Artificial Intelligence

CSIT — 4th Semester — 2080 — Answer Sheet
Subject Code: CSC266

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Define state space graph. Differentiate between A* search and greedy best first search.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What do you mean by unification and lifting? Convert following sentences into FOPL.[4+6] Sushma likes all kinds of practical courses AI and DBMS are practical courses. Any subject anyone practices is practical course. Ruby practices PHP Rita practices everything that Ruby practices Using resolution check whether "Sushma likes PHP" is inferred or not.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Differentiate supervised learning from unsupervised? Discuss How Naive Bayes Model can be used for machine learning? Support your answer with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. How can you define AI from the dimension of behavioral process? When a machine is said to pass Turing Test?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is an agent? How utility agent works? Give an example of utility agent.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is game search? How minimax search is used in game playing? Illustrate with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is semantic network? Given following knowledge base, represent it using semantic network. Subash is a student. All students are person. Person has hair. Ram is a player. All player play game. Game is a physical action. Height of all players is larger than the height of all students. Physical action starts from 7:00 AM and ends at 9:00 AM

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Discuss how genetic algorithm works?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Using your own concepts and construct PEAS framework for following intelligent agents; a. Medicine delivery drone b. Covid Medicine Prescriber

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is machine vision? Describe the components of machine vision.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. How natural language generation differs from natural language understanding? How morphological analysis is done in NLP?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What is constraint satisfaction problem? Illustrate graph coloring problem as a constraint satisfaction problem.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.